

Multiple Choice (10 marks)

1. Which of the following rules regarding the interconversion of fuels are central to the integration of metabolic pathways?
 - (a) Fatty acids can be made from and converted to carbohydrates and amino acids
 - (b) Carbohydrates can be made from and converted to amino acids and fatty acids
 - (c) Fatty acids can be made from but not converted to carbohydrates and amino acids
 - (d) Amino acids can be made from and converted to fatty acids
2. Which nutrient intake is fairly similar for countries in north and south Europe?
 - (a) PUFA
 - (b) MUFA
 - (c) SFA
 - (d) Protein
3. Which of the following lipoproteins is produced via the exogenous lipoprotein pathway?
 - (a) High density lipoproteins
 - (b) Very low density lipoproteins
 - (c) Chylomicrons
 - (d) Low density lipoproteins
4. Fats normally contain triacylglycerols at concentrations of:
 - (a) 75-95%
 - (b) 50- 75%
 - (c) > 95%
 - (d) < 50%
5. In terms of price per calorie, which of the following foods is the cheapest and most nutritious?
 - (a) Apple
 - (b) Slice of bread
 - (c) Cheese burger
 - (d) Broccoli

True / False (10 marks)

Answer True or False

6. True or False: Insulin-like growth factor-1 is associated with an increased risk for prostate cancer.
7. True or False: Endurance exercise training lowers the amount of liver glycogen stored in the body.
8. True or False: Ammonia is the primary nitrogenous compound in urine, which is primarily excreted by the kidneys.
9. True or False: The worldwide prevalence of obesity is around 39%, indicating a significant global health crisis.
10. True or False: It is normal for all babies to lose up to 10% of their birth weight without any cause for concern.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the importance of using screening tools for malnutrition upon hospital admission, and analyze how the MUST tool can effectively identify patients at risk.
12. Discuss the role of vitamin B 12 in the body, particularly in the synthesis of methionine from homocysteine, and explain its significance for overall health.

End of Paper